Status — one line per numbered result

The short answer. `PaperInventory.md` is the long one and stays as the audit trail; this file is the thing to read first.

Measured on `main` after r0047: build **1364 jobs, 0 errors, 0 warnings**, `sorry` **0**, `axiom` **0**, `sorryAx` **0**. Package: 117 modules, 43,481 lines, 2,019 theorems.

Labels

Every paper property carries one of five labels. They answer two questions — **is it stated in Lean, and is it proved** — and the pair matters because **`sorry` can only see the second.**

#	Label	Stated?	Proved?	Meaning
1	`S+P`	yes, as the paper states it	yes	done
2	**`S+H`**	yes, as the paper states it	**no — the proof is open**	a real Lean statement with a hole. `H` is for *hypothesis*: the result is available only conditionally, as a theorem taking the open claim as an argument
3	`S≠`	yes, but **not the paper's statement**	yes	a weakening, an added hypothesis, or a deliberate repair of a printed defect
4	`P`	no — prose only	—	asserted in a docstring, never under the kernel
5	`N`	**no statement at all**	—	nothing in Lean mentions it

Two consequences worth stating once:

- **`sorry` 0 does not mean "everything is proved."** A `sorry` is a hole in a proof someone *started*. Rows 4 and 5 were never started, and row 2's holes are carried as explicit hypotheses rather than as `sorry`, so none of the three appears in a `sorry` count. That is why this file reports rows 8–11 of the counts separately.
- **`S≠` is not automatically a defect.** Five of the 18 are repairs of the paper's own printed errors, which is correct work; 13 are ours.

Counts

#		Count
1	Numbered results in Gunter & Scott	**30, verified** — 16 Theorems, 13 Lemmas, 1 Proposition, 0 Corollaries; numbers 1–30 contiguous, no gap, no repeat
2	**— fully proved**	**25**
3	— partly proved, some part open	3 (results 7, 26, 29)
4	— **refuted as stated** (our transcription or the paper)	2 (results 28, 30)
5	**— all 30 are now stated in the tree**	0 missing
6	Paper properties (numbered conjuncts + prose claims)	239
7	— `S+P` stated and proved	169
8	**— `S+H` stated, proof open**	**12**
9	— `S≠` stated, but not as the paper states it	18 (13 ours, 5 repairs)
10	— `P` + `N` no Lean statement at all	36
11	`Prop`-valued claims nothing proves	**7** (8 strict)

Rows 8 and 11 are the two kinds of unfinished work, and they are different: row 8 is a real Lean statement whose proof is open; row 11 is a `def` naming a claim **nobody attempted**. **`sorry` 0 sees neither** — which is why both are counted here and neither shows up in a build.

The 30 numbered results

`P` proved · `p` partly proved · `R` refuted as stated · – not in the tree

#	Result		Status
1	Theorem 1	P	least fixed point, and below every fixed point
2	**Theorem 2 (Schroder-Bernstein)**	P	*"Let S and T be sets. If* `f : S → T` *and* `g : T → S` *are injections, then there is a bijection* `h : S → T`*"* — printed folio 6. **Genuinely absent until r0048**; now `R48.Agent1.theorem2`. The inventory had mapped it to Mathlib's `Function.Embedding.schroeder_bernstein`, but **that is Zermelo's proof from Knaster–Tarski**, where Gunter & Scott derive it from **Theorem 1** — and `FixedPoint.lean` records that neither implies the other. The new proof follows the paper step for step, via the operator `Y ↦ (T − f*(S)) ∪ f*(g*(Y))`
3	Theorem 3	P	`theorem3`, `theorem3_existsUnique`
4	Lemma 4	P	`NormalSubposet.lean`
5	Lemma 5	P	`FinitaryProjection.lean`
6	Theorem 6	P	`theorem6`
7	Theorem 7	p	sentence 1 proved; sentences 2–3 proved only at the degenerate strength — **every domain has an `EffectivePresentation`** because `Classical.dec` fills its fields. The recursive forms are open: `Theorem7ArrowRecursive`, `Theorem7StrictRecursive`
8	Lemma 8	P	`Product.lean`
9	Lemma 9	P	`lem9_*` — **two printed misprints repaired** (items 3 and 5)
10	Lemma 10	P	7 conjuncts, `lem10_{prod,smash,sum,separated,lift,strict}`
11	Theorem 11	P	`thm11`, with `thm11_converse`
12	Theorem 12	P	`thm12` and the three powerdomains
13	Lemma 13	P	`lem13_smyth`, `lem13_hoare`
14	Theorem 14	P	`thm14`, both directions
15	**Proposition 15**	P	*"A bounded complete domain is bifinite"* — printed folio 31. `prop15` at `Skeleton/Section6.lean:134`, proved since that file was written. **It is a Proposition, not a Theorem or Lemma**, which is the whole reason it read as missing: both instruments that produced the gap searched only for `thm|theorem|lem|lemma`
16	Theorem 16	P	`thm16`, `thm16_positive`
17	Lemma 17	P	10 conjuncts. r0047 **removed `[BoundedComplete β]`** from `lem17_fun` and `lem17_strictFun`
18	Theorem 18	P	`thm18` — closed r0042, `[propext, Classical.choice, Quot.sound]`
19	Lemma 19	P	`lem19`
20	Lemma 20	P	`lem20`
21	Theorem 21	P	`thm21`
22	Theorem 22	P	`thm22`
23	Lemma 23	P	`lem23`
24	Lemma 24	P	`lem24` (Gunter & Scott's; **not** Gunter 1987's Lemma 24, below)
25	Theorem 25	P	`thm25`, `thm25_isUniversal`
26	Theorem 26	p	`thm26` carries an added `hs : ∀ i, 0 < s i` — no nullary operations — which the paper does not assume. The recorded justification, that Theorem 26 is **false** for a signature admitting arity 0, **is not established**: the contradiction is derived from `fst(ψ(x)) = x`, a property of the paper's *construction* and of our own conclusion, whereas the printed statement asks only that `A` "be made isomorphic to a subalgebra" — and two one-point algebras are isomorphic to the same one-point subalgebra $\{F_i\}$. **The argument refutes the paper's proof at arity 0, not its theorem.** So `hs` is a defect of ours, not a repair. To settle it: exhibit a genuine arity-0 counterexample to the *printed* conclusion, or drop `hs` and prove `thm26` without it
27	Theorem 27	P	`thm27`
28	Lemma 28	R	**refuted at generic `U`** (`not_forall_lemma28`, witness `Flat Empty`), and stays false after adding `[Domain U]` and `[BoundedComplete U]`. **Proved at `U`**: `lemma28AtU`, all nine conjuncts. The blocker is `UniversalForBCD U`
29	Theorem 29	p	sentence 1 proved (`thm29`). Sentence 2: `Thm29Second` **refuted** — our transcription dropped the paper's word "domain". `Thm29SecondAtDomains`, the true reading, is **open**

#	Result		Status
30	Lemma 30	`R`	universal closure **refuted** (`not_forall_lemma30`). `Lemma30AtV` is **open**, now at arity 3

Result 15 was never missing — the instruments were. Both the declaration scan and the docstring grep searched only for `thm|theorem|lem|lemma`, and Gunter & Scott number **Propositions in the same sequence**. `scripts/numbered-status.sh` now matches `prop|proposition|cor|corollary` as well; the widening was measured before it was made, and over results 1–30 it adds exactly one hit and no false positive.

Result 2 was genuinely absent, and the two gaps had independent causes: 15 was a grep blind spot, 2 was a real hole. Widening the heading-word set fixes the 15-class defect but **would still have missed 2**.

Results 4, 5 and 8 carry no numbered declaration either, but are each quoted in a module docstring — `NormalSubposet.lean`, `FinitaryProjection.lean`, `Product.lean` — so **a result stated under a descriptive name is the normal case here**, and a missing number is not evidence of a missing proof.

The figure "30 numbered results" is now verified, by extracting every `(Theorem|Lemma|Proposition|Corollary) N` heading from the full text — ASCII headings are unaffected by the Type 3 font defect. Exactly 30, contiguous, no gap and no repeat. The single Proposition among 29 is the entire cause of this round's measurement defect.

What is open, in one table

#	Item	What it needs
1	`StepFunctionsDecidable`	**restate over `consistentEnum`** — the guard `IsCompactElement (ofPairs Q)` is *not* the boundedness test, kernel-checked. Machinery built. Closes 4 claims
2	`Theorem7ArrowRecursive`	item 1, then `Primrec` facts for the `Finset (ℕ × ℕ)` coding
3	`Theorem7StrictRecursive`	item 1, likewise
4	`Thm29Normal`	the `M(f)` tower refutes Theorem 25's hypothesis; the `η` tower is not a fixed point. Three routes named, one is refuting it outright
5	`Thm29SecondAtDomains`	item 4
6	`Lemma30AtV`	item 4, plus `FpImagesBifinite V`
7	`Lem30Arrow`	item 6
8	`PreservesRecursivePresentation`	proved at `fstOp`/`sndOp`; open at the arrow, where it is equivalent to item 2

Items 2–3 and 5–7 are downstream of 1 and 4. Only items 1 and 4 are real work, and item 1 is mechanical.

Additional theories

Results from other authors, formalized here because the paper's proofs need them. None is part of Gunter & Scott's 30.

Jung, *Cartesian Closed Categories of Domains*

#	Result	Status
1	Corollary 1.36	**proved** — `JungCor136`, not by Jung's route (which goes through Prop. 1.22 and a retraction pair) but by indexing below `cap e`
2	Theorem 1.37	**proved at `[Domain D]`** — `R45.Agent5.thm137`. Full generality open
3	Theorem 1.37 for chains	**proved at `[Domain D]`** — `thm137Chains`
4	Proposition 1.22	not needed — the route around it is item 1
5	Theorem 2.1, 2.3	quoted; five printed defects in Jung's write-up recorded

Iwamura and Markowsky

#	Result	Status
1	Iwamura's lemma	**proved** — `exists_chain_directed_cover`. **Never in Mathlib**
2	Markowsky's theorem	**proved** — `hasChainSuprema_iff_hasDirectedSuprema`. **Never in Mathlib**
3	`HasWellOrderedInfima`	**no producer exists** — five occurrences, all consumers. A complete reduction chain with an empty left end

Spreen 2005

#	Result	Status
1	Lemma 5.8	**proved** — `PropertyM.hasOmegaOpBoundsAbove_pair`. This, not Iwamura, is what closed Jung's 1.37 here

Gunter 1987, *Universal Profinite Domains*

#	Result	Status
1	Lemma 24 at `M(A)`	**proved** — `R47.Agent1.lemma24_MPair`. **The first proof anywhere**: Gunter's printed Lemma 24 produces *some* `A⁺` and is not about `M(A)`; the `M(A)` form is a remark with no theorem and no proof
2	`HasNormalRealizations` at `Ainf`	**refuted** — `not_hasNormalRealizations_Ainf`. The route to `Thm29Normal` is sound but its hypothesis is unsatisfiable at this tower
3	Proposition 21, Theorems 22 and 25	used, via the implication `thm29Normal_of_hasNormalRealizations`

Defects in the printed paper

Nine, recorded in `StatementRecovery.md`. Three suspected tenths were checked and turned out to be **our** transcription errors, not the paper's:

#	Suspected	Actually
1	Theorem 26 false at arity 0	refutes the paper's *proof*, not its statement
2	Theorem 29's second sentence false	our transcription dropped "domain" from "any bifinite domain"
3	the step-function enumeration	our guard tests compactness where the paper tests boundedness

Reproducing

```
scripts/counts.sh                  # modules, lines, theorems, sorry
scripts/numbered-status.sh         # declarations per numbered result
scripts/a6-env-scan.sh <out>       # then a6-summarize.py for row 11
scripts/a5-r47-conditional.sh      # row 8, the S+H conditional surface
scripts/compile.sh -r <round>      # build
```